The Core Floor Is Empirical: Closed Mechanism Routes and a Smooth-Core Halo That Passes With Every Input Measured

DRAFT (2026-10-29), not deposited. AI-assisted research programme; not peer reviewed. $\kappa = \frac{1}{2}$ is fitted; c_f is external; the theory is not closed.

Abstract

This note follows PAPER64 (DOI 10.5281/zenodo.23771199). The programme pairs a galaxy law with $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$ ($\kappa = \frac{1}{2}$ fitted) and a gravitating $w \approx 0$ component called *cold dark energy*; this is not matter and not a particle. Its halos obey an empirical core rule:

$$\rho_0 r_0 = \max\big(a_0/2\pi G,\ \eta\,\Sigma_{\rm coll}\big).$$

We report three things. Each comes from frozen lanes: criteria committed before any number, a mutation twin, and identity checks against the committed predecessor.

(i) The floor is not derived, and five routes to deriving it are closed. A local tether, three gated extra pulls on the cold component, phantom-halo tracking, pressure floors and halo reshaping all fail. **(ii) The floor is empirically preferred.** Softening it in low-acceleration galaxies worsens the affected half of SPARC by 43–55%. **(iii) A smooth-cored halo** $\rho_0/(1+x^2)^{3/2}$, with the collapse coefficient re-derived for that shape, passes every graded line on the canonical footing at the assumed stellar fraction. With measured stellar fractions alone, the heaviest-cluster outskirts fail at $Z \approx 2.3$; that miss is X30's truncated gas tail. With every input measured or native (measured X-COP gas slope to turnaround, measured stellar fractions, the programme's own Ω_c/Ω_b), the smooth core passes SPARC, the groups and all 15 cluster keys on the canonical footing at $c_f = 4$. The pass is fragile in the extrapolated gas slope.

(iv) The collapse coefficient measured from our own halos (Section 5): fitting the smooth core directly to native cold-collapse zoom halos gives η at the edge of what the data allow at every mass, with nothing tuned. The full grade is a knife edge: canonical pass at $f_\star = 0.034$, one key just past $|Z| = 2$ at $f_\star = 0.0615$. [PENDING A4] The cluster-mass inputs are still at the coarser grid. This is not evidence for the framework over ΛCDM.

1 The empirical law (as graded)

Canonical footing $a_0 = 9.3603 \times 10^{-11}\,\mathrm{m\,s^{-2}}$, alt 1.1312×10^{-10}; footings are never pooled. The core rule is X43's:

- $\eta = \frac{4}{3B(1)} c_f^{5/4}$, with $c_f = 4$ external;
- $\Sigma_{\rm coll}$ from Lacey–Cole formation on the programme's own $P(k)$;
- the halo normalized to the baryon fair share.

The graded lines are SPARC (175), 63 groups, and 15 cluster lensing keys. TODO: state the relation to PAPER64's basin rule and its native $c_f(M)$ slope.

2 Closed routes to deriving the floor (negative results)

Lane	Mechanism	Failure
X47	local a_0 tether	a cored profile has $g < g_T$ everywhere: no floor (latching: everything is dust by $z \sim 40$–700)
X65	AQUAL pull gated by stream sign Q	lifts every halo ×3–5; no plateau; max() broken
X66	core-only boost to $g_\sigma = a_0/4$	barely engages; heats cores (R 0.21→0.15)
X67	tidal $(\bar\rho/\rho)^2$ screening	softer X65; boost reaches 50–100 r_0
–	phantom tracking (Milgrom 2009)	gives a *ceiling* $\lesssim a_0/2\pi G$, not a floor [PROVISIONAL transcription]
–	Λ-set pressure / phase-space floor	can only lower $\rho_0 r_0$
X68	split inner/outer scale at fixed core Σ	no solution: 2.5–5.5× too heavy (mass budget is zero-sum)
–	correlated matter beyond $r_{\rm ta}$	$r_{\rm ta} \approx 7\,$Mpc comoving; mean-subtracted column $< 1\,M_\odot\,\mathrm{pc^{-2}}$ at 2 Mpc; $\Delta\Sigma_{\rm ext} < 0$
–	curvature / Einstein–Hilbert restatements of $\kappa = \frac{1}{2}$	assume the gradient scale $\sqrt{G\rho_{\rm DE}}/c$; restatement, not derivation (cf. DSF-21)

Scope note: X65–X67 used a 1D shell harness in which each shell keeps its angular momentum. Core sizes there are partly set by that angular momentum. The harness diagnoses global failure modes; it cannot certify a floor. TODO: cite `X65_X67_POSTMORTEM.md`.

3 The floor is preferred by SPARC

X69_soft_floor_sparc replaced the floor by $\Sigma_M F(X_0)$, with $F = 4X_0/\sqrt{1+16X_0^2}$ and X_0 the peak baryonic acceleration over a_0.

- Low-X_0 galaxies are 93 of 175 (median X_0 0.29), not a small tail.
- Their rms worsens by 54.5% (canonical) and 44.1% (alt). The SPARC line fails, with median $\Upsilon > 1$.

An independent peer lane (X70_soft_floor_x0) agrees.

4 A smooth-cored halo

X70_smooth_core_shape replaces Burkert with $\rho_0/(1+x^2)^{3/2}$, which has zero central slope and the same r^{-3} tail. It re-derives η from the same infall mass match: $B_S(1) = 0.69707$, so $\eta_S = 10.82$.

- **Canonical:** SPARC rms 0.0713 (Υ 0.56); all 15 cluster keys $|Z| \leq 1.88$; groups $Z = +1.94$ (CONSISTENT).
- **Alt:** cluster keys and groups pass; the SPARC line fails on $\Upsilon = 0.47 < 0.50$.

X71 stress test (measured stellar fractions, uniform):

- G-OUT-M3: $Z = +2.25$ ($f_\star = 0.0615$) and $+2.33$ ($f_\star = 0.034$) in the $h = 0.672$ cell.
- Everything else still passes.
- Burkert at the same $f_\star$: $+2.75$ and $+2.83$, and the groups fail.

X73_smooth_core_all_measured (frozen before any number) replaces X30's truncated "P" gas tail with the measured X-COP slope $s = 2.777$ continued to $r_{\rm ta}$, and Planck's Ω_c/Ω_b with the programme's own:

- Canonical, primary: all 15 cluster keys pass; G-OUT-M3 $Z = -0.75$ ($f_\star = 0.034$) and -0.89 ($f_\star = 0.0615$); groups $Z = +1.94$; SPARC rms 0.0713.
- X71's miss was the truncated gas tail, not the stellar fraction. With the measured tail, X43's $f_\star = 0.20$ itself over-predicts (G-OUT-all $Z = -2.28$).
- **Fragile:** at the 16th-percentile slope G-OUT-M3 reaches $Z = -2.01$ (over-prediction); at the 84th, $+1.46$. The outer keys measure the gas beyond $\approx 2R_{500}$, which is extrapolated.
- Alt: clusters behave the same; X70's SPARC Υ failure stands.

Look-elsewhere: the shape was proposed after X43's misses were known. Only this one alternative was frozen and run.

5 The collapse coefficient from native halos

The grader uses only $\rho_0 r_0 = \eta\Sigma_{\rm coll}$; c_f enters through X43's map $\eta \propto c_f^{5/4}$, which assumes the infall law $M \propto r^{0.75}$. X77_direct_core_surface_density (frozen before any number) bypasses both. It fits the grader's own smooth-core shape to every native X48 zoom halo (collisionless cold collapse of an isolated constrained peak; X48's centre, $r_{\rm vir}$, fit range and objective unchanged) and reads

$$\eta_{\rm native} \equiv \frac{\rho_0 r_0}{18\pi^2 \rho_m r_{\rm vir}} = \frac{4a^2}{3B_S(a)}, \qquad a = r_{\rm vir}/r_0.$$

The limits below are from committed lanes: the SPARC Υ floor at 10^{12}, and G-IN-Mall at cluster mass.

mass, grid	$\eta_{\rm native}$ (nine-seed median)	limit	grade
10^{12}, ng 64	20.83	21.2	inside
10^{13}, ng 48	13.79	14.3	inside
10^{14}, ng 48	14.39	14.3	over $\times 1.01$

- **The infall map is sound; the r_{-2} crossing was not.** A synthetic NFW halo with $c = 4$ gives $\eta_{\rm native} = 10.34$, against the rule's 10.82. At cluster mass the direct η corresponds to $c_f \approx 5$, where the slope-crossing estimator read 7–11. The apparent factor-two over-concentration of native halos was an estimator artefact.
- **Full grade with the native η** (X73 configuration, canonical): at $f_\star = 0.034$ every line passes (worst cluster key G-IN-Mall $Z = -1.99$); at $f_\star = 0.0615$ G-IN-Mall fails at -2.04 (too dense) and everything else passes. Alt fails the SPARC line ($\Upsilon = 0.45$), as at every c_f.
- Resolution: at 10^{12}, $\eta_{\rm native}$ shifts +14.5% from ng 48 to ng 64.

Reading: halos made by the programme's own cold collapse land on the boundary of what the data allow, at every mass, with nothing tuned. [PENDING A4] The ng 64 cluster zooms replace the ng 48 entries. If cluster η rises like the 10^{12} value, the clusters come out too dense by about 15%, and c_f remains a calibration. The X74 and X72 regrades (frozen before A4) are reported as frozen.

6 What is fitted, posited or failing

- $\kappa = \frac{1}{2}$: fitted. No proposed normalization (de Sitter $R = 4\Lambda$, "$\sqrt{8\pi/32\pi}$") derives it without assuming it.
- $c_f = 4$: external in the rule; native halos give an equivalent $c_f \approx 5$–7 at the window edge (Section 5) [PENDING A4].
- The floor $a_0/2\pi G$: empirical; every mechanism tested is closed (Section 2).
- Fragile: heavy-cluster outskirts, which hinge on the gas slope beyond $2R_{500}$ (a measurement, not a model, decides it).
- Failing: the SPARC Υ line on alt; IC 1613 (PAPER64 v1.2). Not regraded here: KiDS lensing, the dwarfs.